Joel Mathewjames

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.08
Greets Public School	Class XII (C.B.S.E)	2018	82.00%
Greets Public School	Class X (C.B.S.E)	2020	80.00%

EXPERIENCE

Trainee - Medical Trust Hospital

Spearheaded administrative workflow, Directed equipment setup and vendor communications for a 200-bed ICU expansion, while concurrently coordinating layout planning for a multi-modality super-specialty radiology unit.

Biomedical Apprentice - Regional Cancer Centre (RCC), Thiruvananthapuram

Maintained high-value radiology infrastructure across a 500+ bed oncology center, delivering dedicated clinical engineering support for complex da Vinci robotic surgical systems

POSITIONS OF RESPONSIBILITY

Design Lead & Event Media Coordinator (2026)

Spearheaded visual branding and promotional media for major academic symposia, directing creative assets for the India AI Impact Summit (300+ attendees) and Cardiovascular Research Convergence (200+ clinicians and researchers)

Class Representative (CR) - M.Tech Biomedical Engineering

Elected to manage batch-wide academic coordination, successfully streamlining administrative communications between the student cohort and engineering faculty.

Postgraduate (PG) Mentor - Sunshine Club

Academic and peer mentor guiding incoming postgraduate students through campus onboarding, while organizing student welfare events impacting over 500 students.

SKILLS

AI / ML for Biomedical Applications

Machine learning and deep learning architecture design (including CNNs and autoencoders) focused on time-series classification, automated arrhythmia detection, and physiological signal analysis

Programming & Libraries:Python - Advanced utilization of NumPy and SciPy for numerical computing, NeuroKit2 and SciPy.signal for biomedical signal processing, Matplotlib for data visualization, and scikit-learn / PyTorch for AI/ML pipelines

Hardware, Embedded Systems & Prototyping: Practical hands-on expertise in PCB layout and AFE circuit design (KiCad), embedded firmware and telemetry (ESP32, Raspberry Pi, I2C/SPI/UART), and 3D CAD modeling (Fusion 360, SolidWorks) with FDM 3D printing for custom sensor molds and wearable enclosures

PROJECTS

Novel Carbon Nanocomposite Electrodes for Biopotential Acquisition

Engineered novel flexible carbon-based conductive composites to fabricate soft dry electrodes, optimizing signal

acquisition for long-term, gel-free wearable biosensors

Ambulatory Cardiac Telemetry Band via Custom Carbon Electrodes

Designed and prototyped a single-arm, single-lead wearable ECG band utilizing custom carbon dry electrodes for continuous, high-fidelity ambulatory cardiac telemetry

Early Sepsis Prediction in ICU using MIMIC-III Clinical Data

Analyzed complex, multidimensional electronic health records from the MIMIC-III dataset to architect an AI-driven predictive pipeline for early sepsis detection in intensive care settings

Flexible Piezoresistive Tactile Sensors for Dynamic Force Mapping

Developed highly responsive, flexible piezoresistive sensors using carbon nanocomposites to accurately map tactile biomechanical data for wearable human-machine interfaces.

Dry EEG Electrode Design & Optimization

Engineered dry biopotential electrode geometries to optimize scalp-electrode contact impedance, reducing setup time while maintaining high signal fidelity

RELEVANT COURSES

Accident and Emergency Informatics

Explored AI model architectures in healthcare, Electronic Health Record (EHR) data processing, and clinical informatics applications.

Artificial Intelligence in Healthcare -- Regulation & Standards

Examined regulatory frameworks for medical AI, including FDA Software as a Medical Device (SaMD) guidelines, Good Machine Learning Practice (GMLP), and the EU AI Act

EXTRACURRICULAR

Sports & Athletics

Athlete on the institutional track & field team (Shot Put & Discus Throw); represented the institute at the Inter-IIT Sports Meet.

Photography & Videography

Experienced in visual storytelling, cinematography, video editing, and digital media production.

Graphic Design & Creative Media

Proficient in promotional design, poster creation, and visual branding assets for events and symposia